Workforce Attrition Analysis and Employee Retention Insights

Overview

This report summarizes foundational SQL-based queries executed on a representative employee attrition dataset. The analysis applies standard aggregation and sorting techniques to key metrics, Age, tenure (YearsAtCompany), prior experience (NumCompaniesWorked), department/education field proxies, gender, marital status, and attrition status to generate actionable insights.

Tools and methodology

All queries were executed using Microsoft SQL Server. No advanced ETL or programming languages were required for this foundational layer, ensuring accessibility and reproducibility by internal analytics teams.

Queries were run directly against a single table named attrition, with:

- Full result sets captured for sorting tasks (ORDER BY).
- Aggregated outputs for SUM, AVG, COUNT, MIN, and MAX functions.
- WHERE clauses for filtering (e.g., by Department, Gender, EducationField, Attrition = 'Yes').
- GROUP BY for departmental breakdowns.

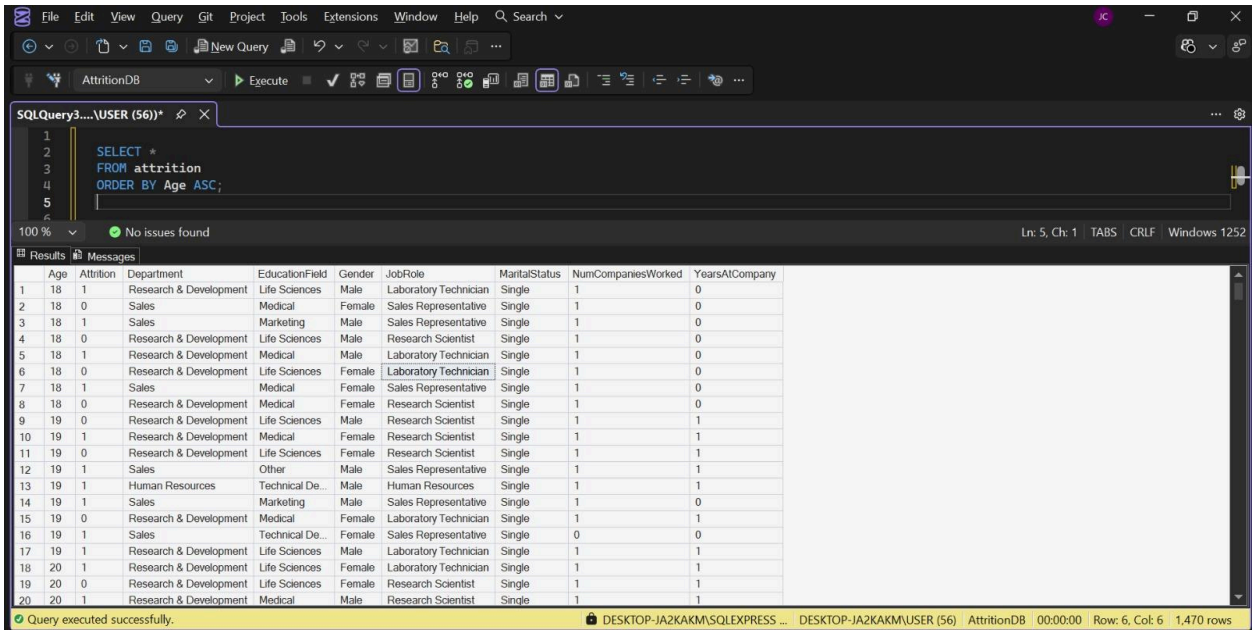
Screenshots of each query (including the executed code and result grid) were documented to verify correctness and enable auditability which is critical for shareholder assurance on data-driven decision-making.

SQL QUERIES

1. ORDER BY (Ascending)

1. Retrieve all records from the attrition table ordered by age in ascending order.

```
SELECT *

FROM attrition

ORDER BY Age ASC;
```

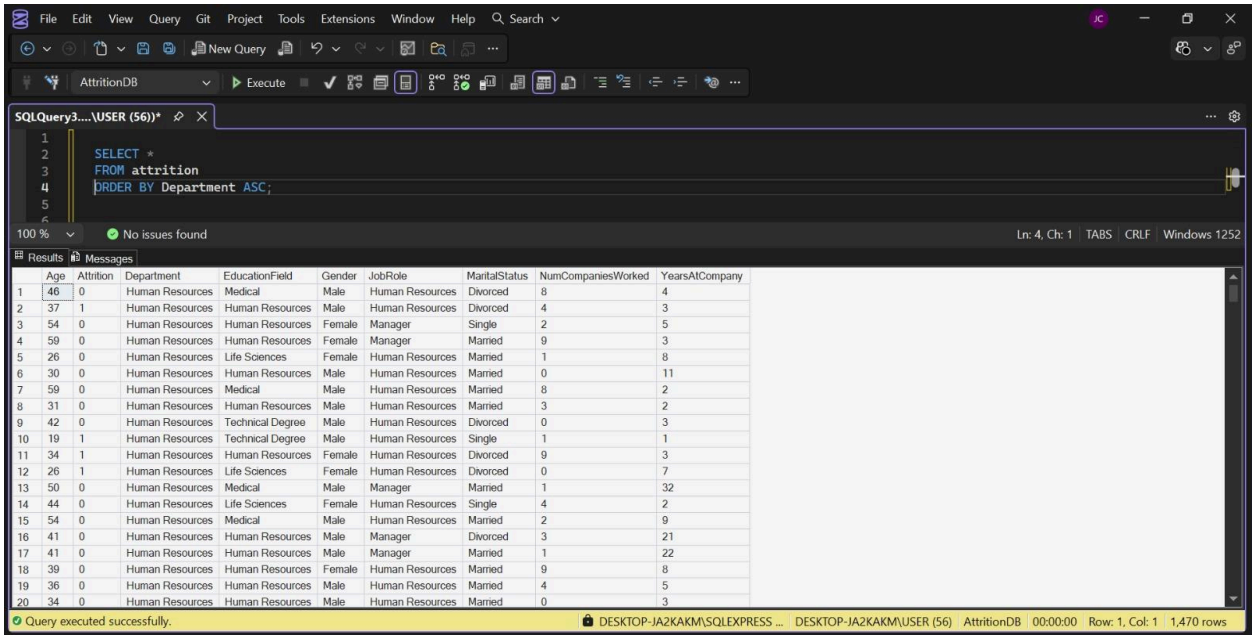


	Age	Attrition	Department	EducationField	Gender	JobRole	MaritalStatus	NumCompaniesWorked	YearsAtCompany
1	18	1	Research & Development	Life Sciences	Male	Laboratory Technician	Single	1	0
2	18	0	Sales	Medical	Female	Sales Representative	Single	1	0
3	18	1	Sales	Marketing	Male	Sales Representative	Single	1	0
4	18	0	Research & Development	Life Sciences	Male	Research Scientist	Single	1	0
5	18	1	Research & Development	Medical	Male	Laboratory Technician	Single	1	0
6	18	0	Research & Development	Life Sciences	Female	Laboratory Technician	Single	1	0
7	18	1	Sales	Medical	Female	Sales Representative	Single	1	0
8	18	0	Research & Development	Medical	Female	Research Scientist	Single	1	0
9	19	0	Research & Development	Life Sciences	Male	Research Scientist	Single	1	1
10	19	1	Research & Development	Medical	Female	Research Scientist	Single	1	1
11	19	0	Research & Development	Life Sciences	Female	Research Scientist	Single	1	1
12	19	1	Sales	Other	Male	Sales Representative	Single	1	1
13	19	1	Human Resources	Technical De...	Male	Human Resources	Single	1	1
14	19	1	Sales	Marketing	Male	Sales Representative	Single	1	0
15	19	0	Research & Development	Medical	Female	Laboratory Technician	Single	1	1
16	19	1	Sales	Technical De...	Female	Sales Representative	Single	0	0
17	19	1	Research & Development	Life Sciences	Male	Laboratory Technician	Single	1	1
18	20	1	Research & Development	Life Sciences	Female	Laboratory Technician	Single	1	1
19	20	0	Research & Development	Life Sciences	Female	Research Scientist	Single	1	1
20	20	1	Research & Development	Medical	Male	Research Scientist	Single	1	1

Summary: This lists employees from youngest to oldest. The youngest employees appear first, useful for analyzing entry-level staff patterns.

2. Display the attrition table, ordering the data alphabetically by department in ascending order.

```
SELECT *

FROM attrition

ORDER BY Department ASC;
```

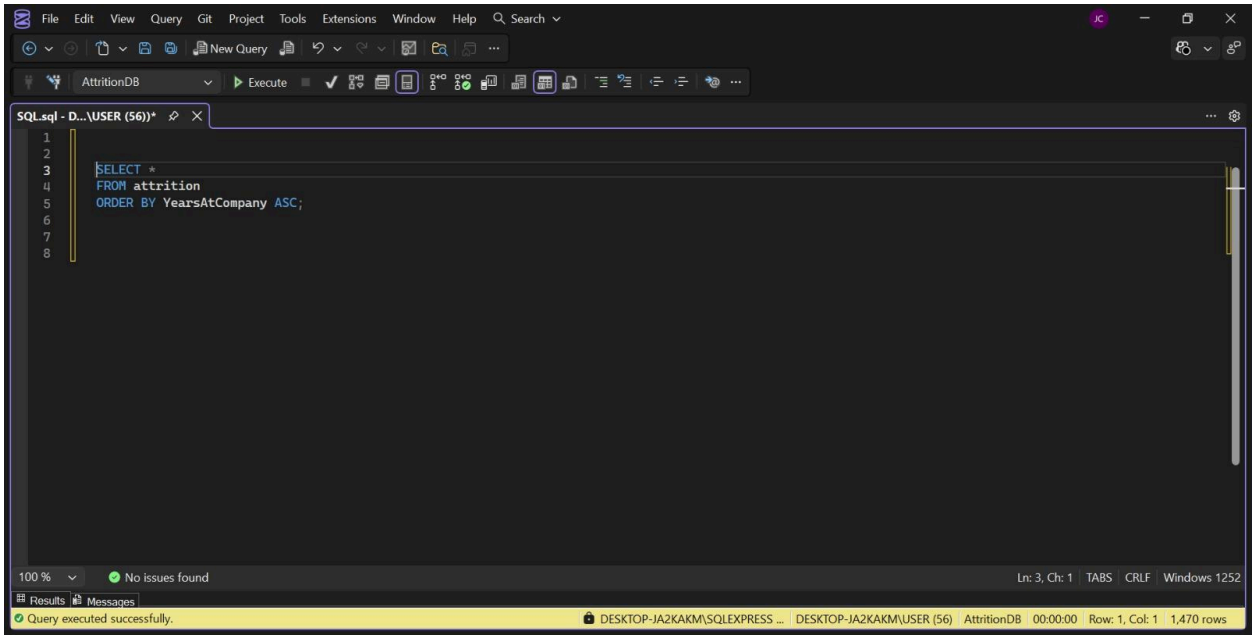


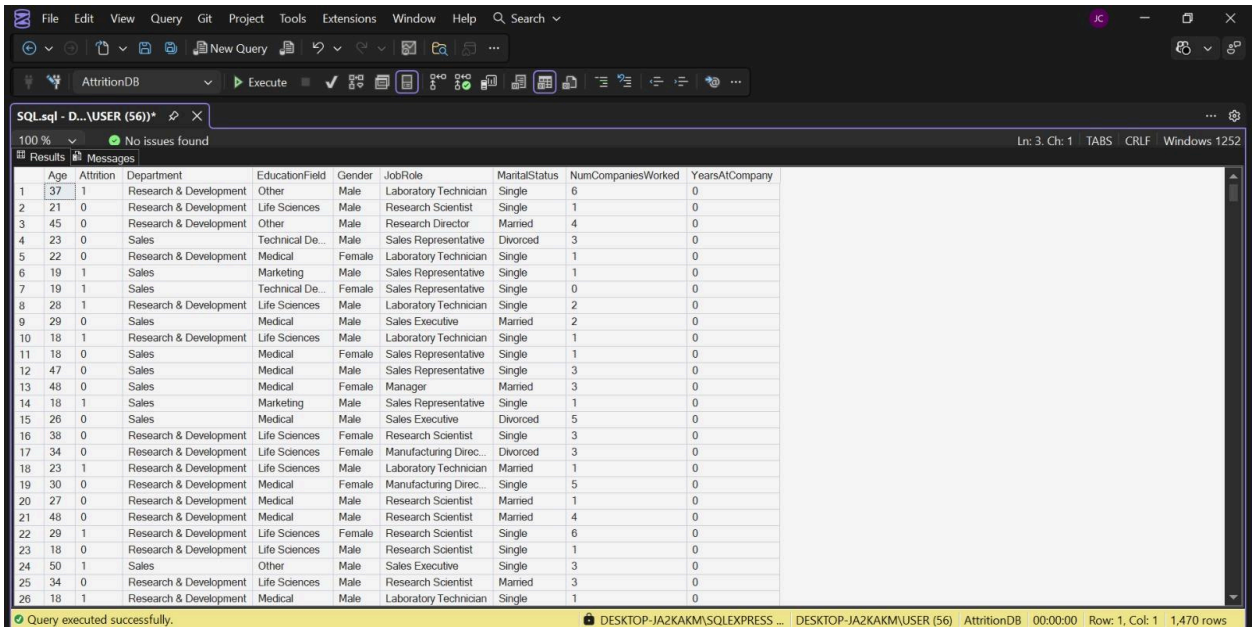
	Age	Attrition	Department	EducationField	Gender	JobRole	MaritalStatus	NumCompaniesWorked	YearsAtCompany
1	46	0	Human Resources	Medical	Male	Human Resources	Divorced	8	4
2	37	1	Human Resources	Human Resources	Male	Human Resources	Divorced	4	3
3	54	0	Human Resources	Human Resources	Female	Manager	Single	2	5
4	59	0	Human Resources	Human Resources	Female	Manager	Married	9	3
5	26	0	Human Resources	Life Sciences	Female	Human Resources	Married	1	8
6	30	0	Human Resources	Human Resources	Male	Human Resources	Married	0	11
7	59	0	Human Resources	Medical	Male	Human Resources	Married	8	2
8	31	0	Human Resources	Human Resources	Male	Human Resources	Married	3	2
9	42	0	Human Resources	Technical Degree	Male	Human Resources	Divorced	0	3
10	19	1	Human Resources	Technical Degree	Male	Human Resources	Single	1	1
11	34	1	Human Resources	Human Resources	Female	Human Resources	Divorced	9	3
12	26	1	Human Resources	Life Sciences	Female	Human Resources	Divorced	0	7
13	50	0	Human Resources	Medical	Male	Manager	Married	1	32
14	44	0	Human Resources	Life Sciences	Female	Human Resources	Single	4	2
15	54	0	Human Resources	Medical	Male	Human Resources	Married	2	9
16	41	0	Human Resources	Human Resources	Male	Manager	Divorced	3	21
17	41	0	Human Resources	Human Resources	Male	Manager	Married	1	22
18	39	0	Human Resources	Human Resources	Female	Human Resources	Married	9	8
19	36	0	Human Resources	Human Resources	Male	Human Resources	Married	4	5
20	34	0	Human Resources	Human Resources	Male	Human Resources	Married	0	3

Summary: Departments are ordered alphabetically (Human Resources first, then Research & Development, Sales last). Helps view employees grouped by department.

3. Show the attrition table sorted by the number of years at the company in ascending order.

```
SELECT *

FROM attrition

ORDER BY YearsAtCompany ASC;
```





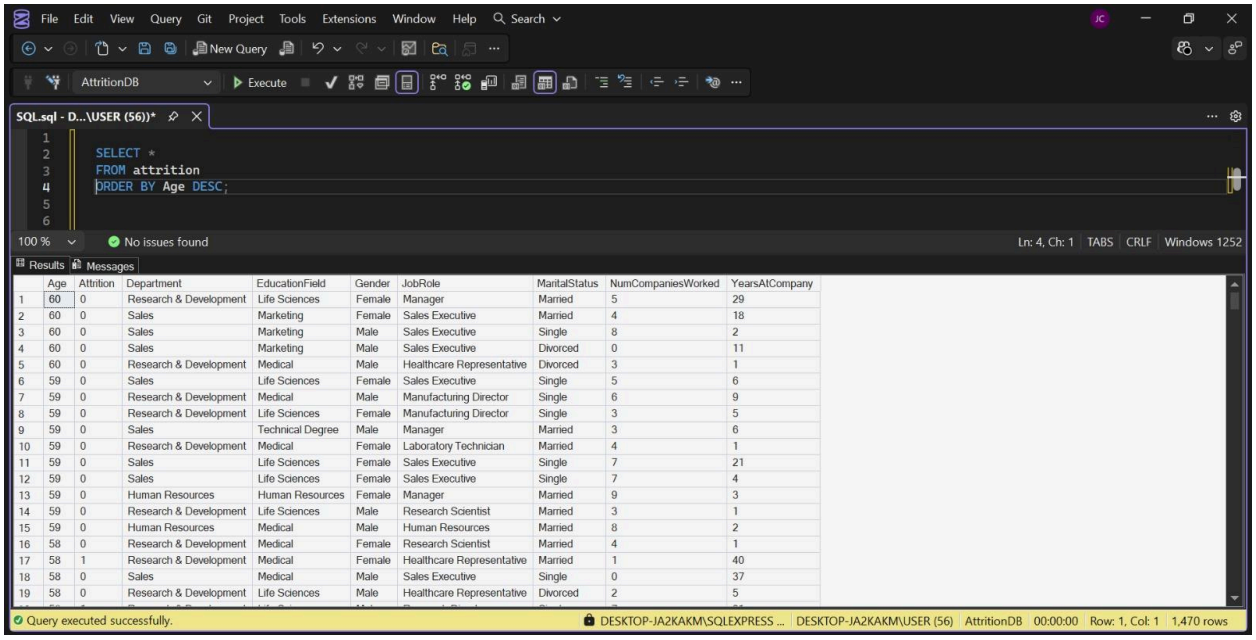
	Age	Attrition	Department	EducationField	Gender	JobRole	MaritalStatus	NumCompaniesWorked	YearsAtCompany
1	37	1	Research & Development	Other	Male	Laboratory Technician	Single	6	0
2	21	0	Research & Development	Life Sciences	Male	Research Scientist	Single	1	0
3	45	0	Research & Development	Other	Male	Research Director	Married	4	0
4	23	0	Sales	Technical De...	Male	Sales Representative	Divorced	3	0
5	22	0	Research & Development	Medical	Female	Laboratory Technician	Single	1	0
6	19	1	Sales	Marketing	Male	Sales Representative	Single	1	0
7	19	1	Sales	Technical De...	Female	Sales Representative	Single	0	0
8	28	1	Research & Development	Life Sciences	Male	Laboratory Technician	Single	2	0
9	29	0	Sales	Medical	Male	Sales Executive	Married	2	0
10	18	1	Research & Development	Life Sciences	Male	Laboratory Technician	Single	1	0
11	18	0	Sales	Medical	Female	Sales Representative	Single	1	0
12	47	0	Sales	Medical	Male	Sales Representative	Single	3	0
13	48	0	Sales	Medical	Female	Manager	Married	3	0
14	18	1	Sales	Marketing	Male	Sales Representative	Single	1	0
15	26	0	Sales	Medical	Male	Sales Executive	Divorced	5	0
16	38	0	Research & Development	Life Sciences	Female	Research Scientist	Single	3	0
17	34	0	Research & Development	Life Sciences	Female	Manufacturing Direc...	Divorced	3	0
18	23	1	Research & Development	Life Sciences	Male	Laboratory Technician	Married	1	0
19	30	0	Research & Development	Medical	Female	Manufacturing Direc...	Single	5	0
20	27	0	Research & Development	Medical	Male	Research Scientist	Married	1	0
21	48	0	Research & Development	Medical	Male	Research Scientist	Married	4	0
22	29	1	Research & Development	Life Sciences	Female	Research Scientist	Single	6	0
23	18	0	Research & Development	Life Sciences	Male	Research Scientist	Single	1	0
24	50	1	Sales	Other	Male	Sales Executive	Single	3	0
25	34	0	Research & Development	Life Sciences	Male	Research Scientist	Married	3	0
26	18	1	Research & Development	Medical	Male	Laboratory Technician	Single	1	0

Summary: Newest employees (0–1 years) appear first; longer-tenured ones at the bottom. Useful for tenure-based analysis.

2. ORDER BY (Descending)

1. Retrieve all records from the attrition table ordered by age in descending order.

```
SELECT *
FROM attrition
ORDER BY Age DESC;
```

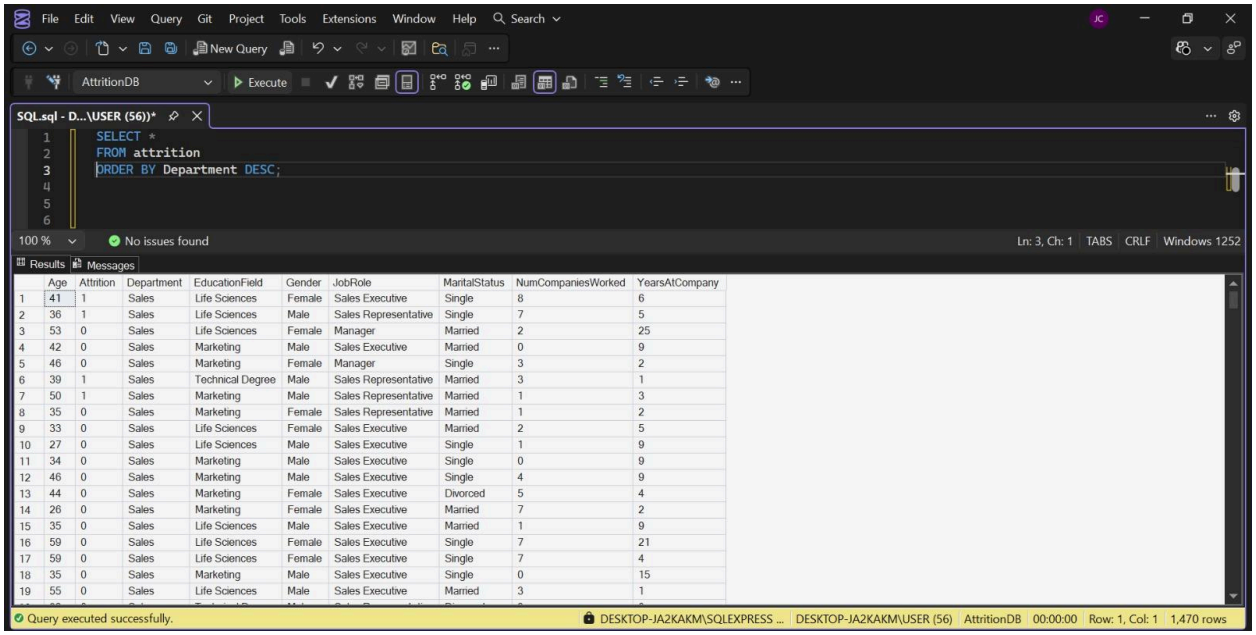


	Age	Attrition	Department	EducationField	Gender	JobRole	MaritalStatus	NumCompaniesWorked	YearsAtCompany
1	60	0	Research & Development	Life Sciences	Female	Manager	Married	5	29
2	60	0	Sales	Marketing	Female	Sales Executive	Married	4	18
3	60	0	Sales	Marketing	Male	Sales Executive	Single	8	2
4	60	0	Sales	Marketing	Male	Sales Executive	Divorced	0	11
5	60	0	Research & Development	Medical	Male	Healthcare Representative	Divorced	3	1
6	59	0	Sales	Life Sciences	Female	Sales Executive	Single	5	6
7	59	0	Research & Development	Medical	Male	Manufacturing Director	Single	6	9
8	59	0	Research & Development	Life Sciences	Female	Manufacturing Director	Single	3	5
9	59	0	Sales	Technical Degree	Male	Manager	Married	3	6
10	59	0	Research & Development	Medical	Female	Laboratory Technician	Married	4	1
11	59	0	Sales	Life Sciences	Female	Sales Executive	Single	7	21
12	59	0	Sales	Life Sciences	Female	Sales Executive	Single	7	4
13	59	0	Human Resources	Human Resources	Female	Manager	Married	9	3
14	59	0	Research & Development	Life Sciences	Male	Research Scientist	Married	3	1
15	59	0	Human Resources	Medical	Male	Human Resources	Married	8	2
16	58	0	Research & Development	Medical	Female	Research Scientist	Married	4	1
17	58	1	Research & Development	Medical	Female	Healthcare Representative	Married	1	40
18	58	0	Sales	Medical	Male	Sales Executive	Single	0	37
19	58	0	Research & Development	Life Sciences	Male	Healthcare Representative	Divorced	2	5

Summary: Oldest employees listed first. Reveals senior staff distribution.

2. Display the attrition table, ordering the data alphabetically by department in descending order.

```
SELECT *
FROM attrition
ORDER BY Department DESC;
```

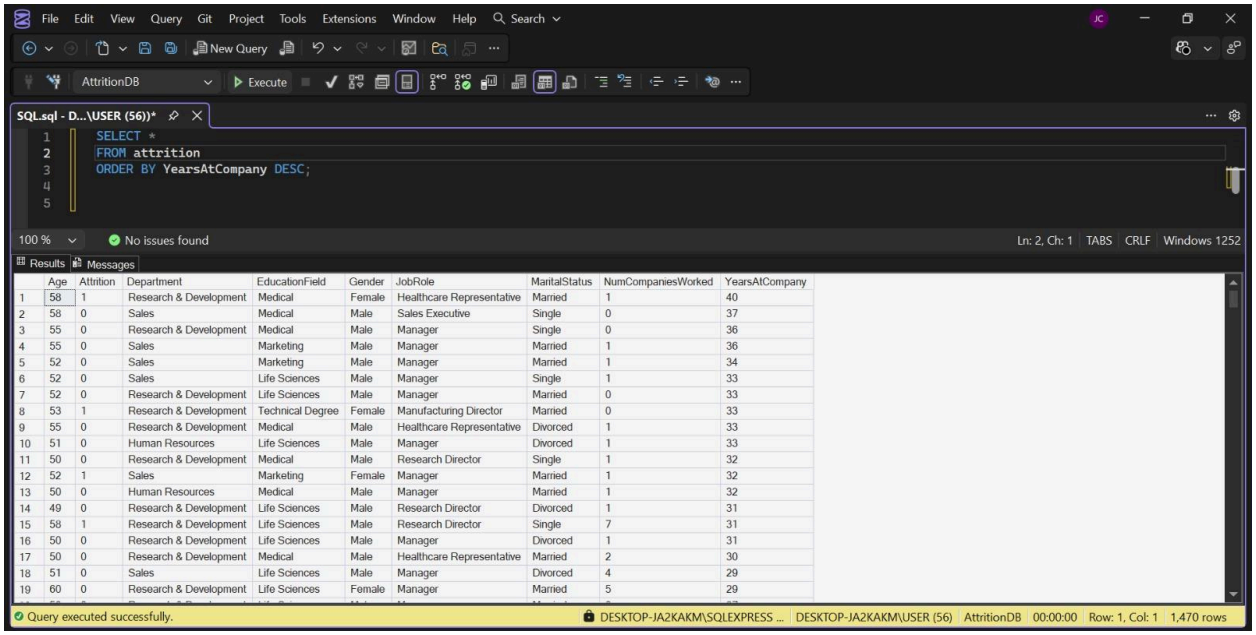


	Age	Attrition	Department	EducationField	Gender	JobRole	MaritalStatus	NumCompaniesWorked	YearsAtCompany
1	41	1	Sales	Life Sciences	Female	Sales Executive	Single	8	6
2	36	1	Sales	Life Sciences	Male	Sales Representative	Single	7	5
3	53	0	Sales	Life Sciences	Female	Manager	Married	2	25
4	42	0	Sales	Marketing	Male	Sales Executive	Married	0	9
5	46	0	Sales	Marketing	Female	Manager	Single	3	2
6	39	1	Sales	Technical Degree	Male	Sales Representative	Married	3	1
7	50	1	Sales	Marketing	Male	Sales Representative	Married	1	3
8	35	0	Sales	Marketing	Female	Sales Representative	Married	1	2
9	33	0	Sales	Life Sciences	Female	Sales Executive	Married	2	5
10	27	0	Sales	Life Sciences	Male	Sales Executive	Single	1	9
11	34	0	Sales	Marketing	Male	Sales Executive	Single	0	9
12	46	0	Sales	Marketing	Male	Sales Executive	Single	4	9
13	44	0	Sales	Marketing	Female	Sales Executive	Divorced	5	4
14	26	0	Sales	Marketing	Female	Sales Executive	Married	7	2
15	35	0	Sales	Life Sciences	Male	Sales Executive	Married	1	9
16	59	0	Sales	Life Sciences	Female	Sales Executive	Single	7	21
17	59	0	Sales	Life Sciences	Female	Sales Executive	Single	7	4
18	35	0	Sales	Marketing	Male	Sales Executive	Single	0	15
19	55	0	Sales	Life Sciences	Male	Sales Executive	Married	3	1

Summary: Reverse alphabetical order (Sales first, then Research & Development, Human Resources last).

3. Show the attrition table sorted by the number of years at the company in descending order.

```
SELECT *

FROM attrition

ORDER BY YearsAtCompany DESC;
```



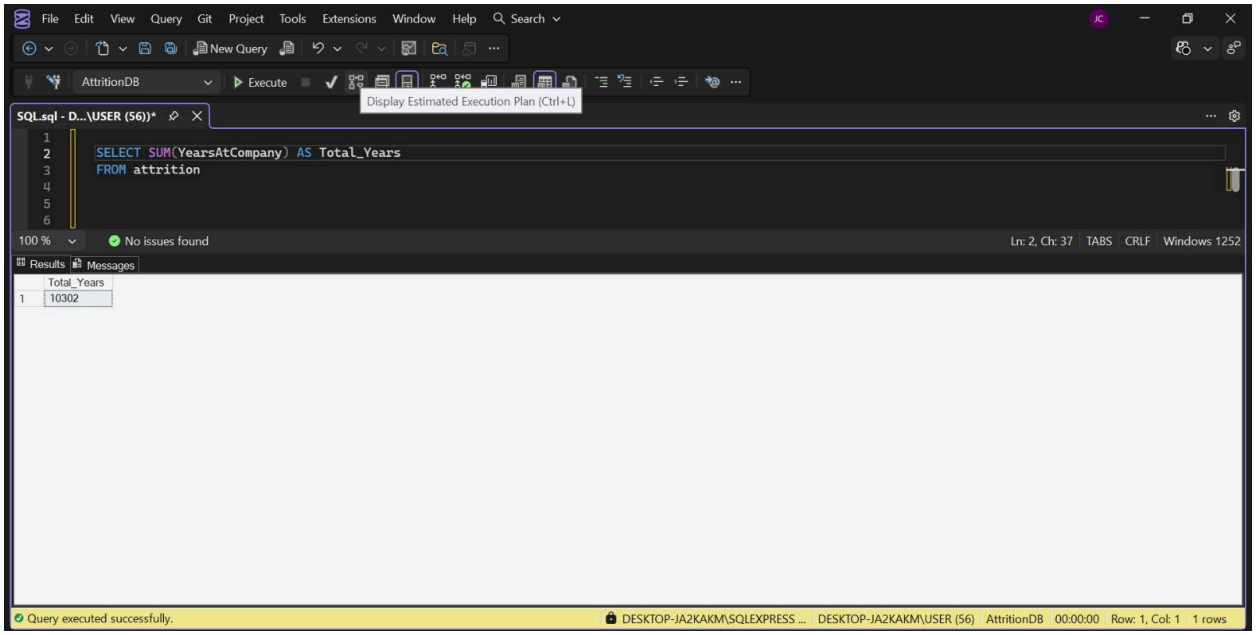
	Age	Attrition	Department	EducationField	Gender	JobRole	MaritalStatus	NumCompaniesWorked	YearsAtCompany
1	58	1	Research & Development	Medical	Female	Healthcare Representative	Married	1	40
2	58	0	Sales	Medical	Male	Sales Executive	Single	0	37
3	55	0	Research & Development	Medical	Male	Manager	Single	0	36
4	55	0	Sales	Marketing	Male	Manager	Married	1	36
5	52	0	Sales	Marketing	Male	Manager	Married	1	34
6	52	0	Sales	Life Sciences	Male	Manager	Single	1	33
7	52	0	Research & Development	Life Sciences	Male	Manager	Married	0	33
8	53	1	Research & Development	Technical Degree	Female	Manufacturing Director	Married	0	33
9	55	0	Research & Development	Medical	Male	Healthcare Representative	Divorced	1	33
10	51	0	Human Resources	Life Sciences	Male	Manager	Divorced	1	33
11	50	0	Research & Development	Medical	Male	Research Director	Single	1	32
12	52	1	Sales	Marketing	Female	Manager	Married	1	32
13	50	0	Human Resources	Medical	Male	Manager	Married	1	32
14	49	0	Research & Development	Life Sciences	Male	Research Director	Divorced	1	31
15	58	1	Research & Development	Life Sciences	Male	Research Director	Single	7	31
16	50	0	Research & Development	Life Sciences	Male	Manager	Divorced	1	31
17	50	0	Research & Development	Medical	Male	Healthcare Representative	Married	2	30
18	51	0	Sales	Life Sciences	Male	Manager	Divorced	4	29
19	60	0	Research & Development	Life Sciences	Female	Manager	Married	5	29

Summary: Longest-serving employees first (high tenure at top).

3. SUM

1. Calculate the total number of years at the company for all employees.

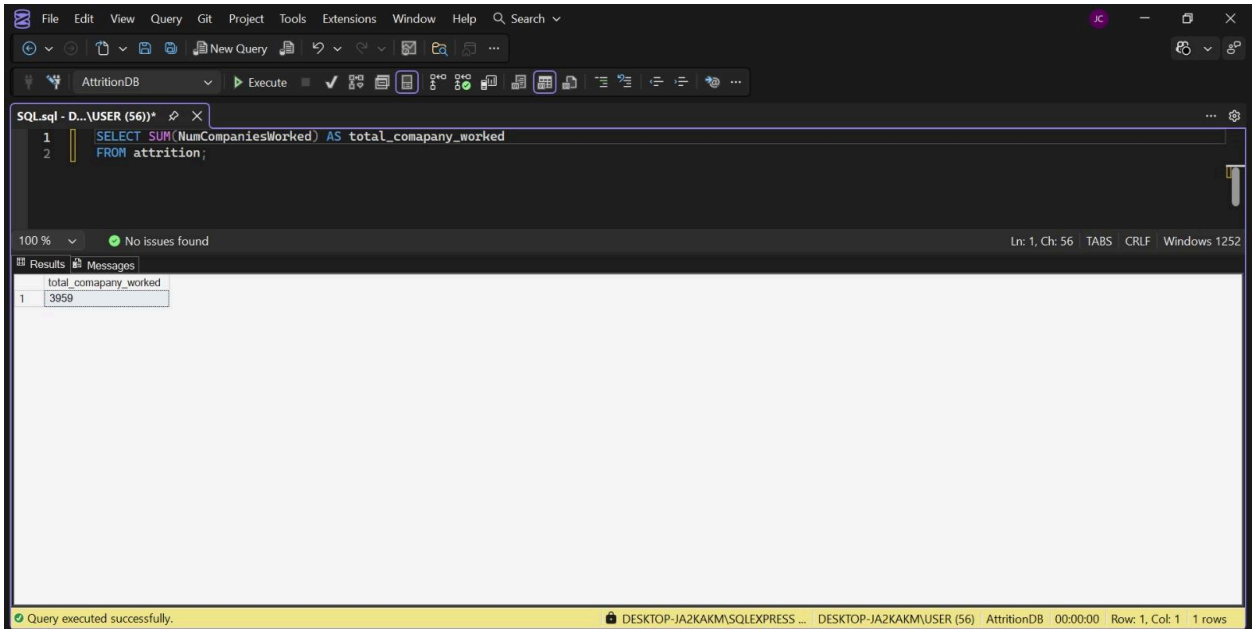
```
SELECT SUM(YearsAtCompany) AS Total_Years
FROM attrition;
```



Summary: Represents cumulative experience within the company across the workforce.

2. Find the sum of the number of companies worked at by all employees.

```
SELECT SUM(NumCompaniesWorked) AS total_comapany_worked
FROM attrition;
```

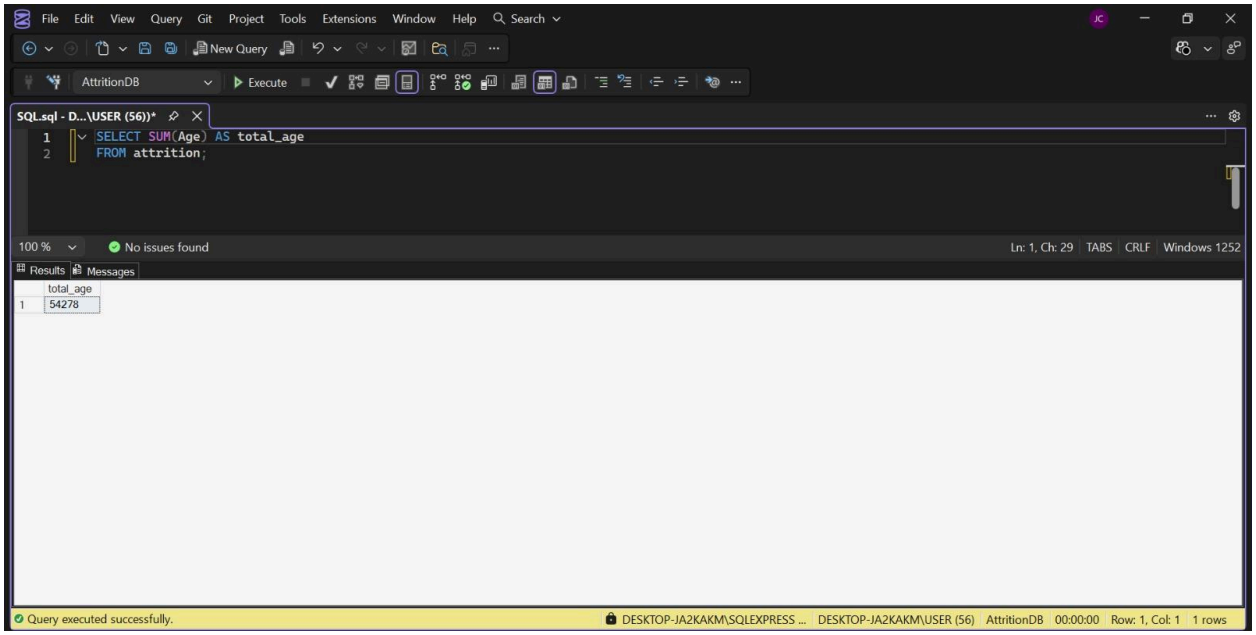


Summary: Total prior companies across all employees (indicates job-hopping history).

3. Calculate the total age of all employees in the attrition table.

SELECT SUM(Age) AS total_age

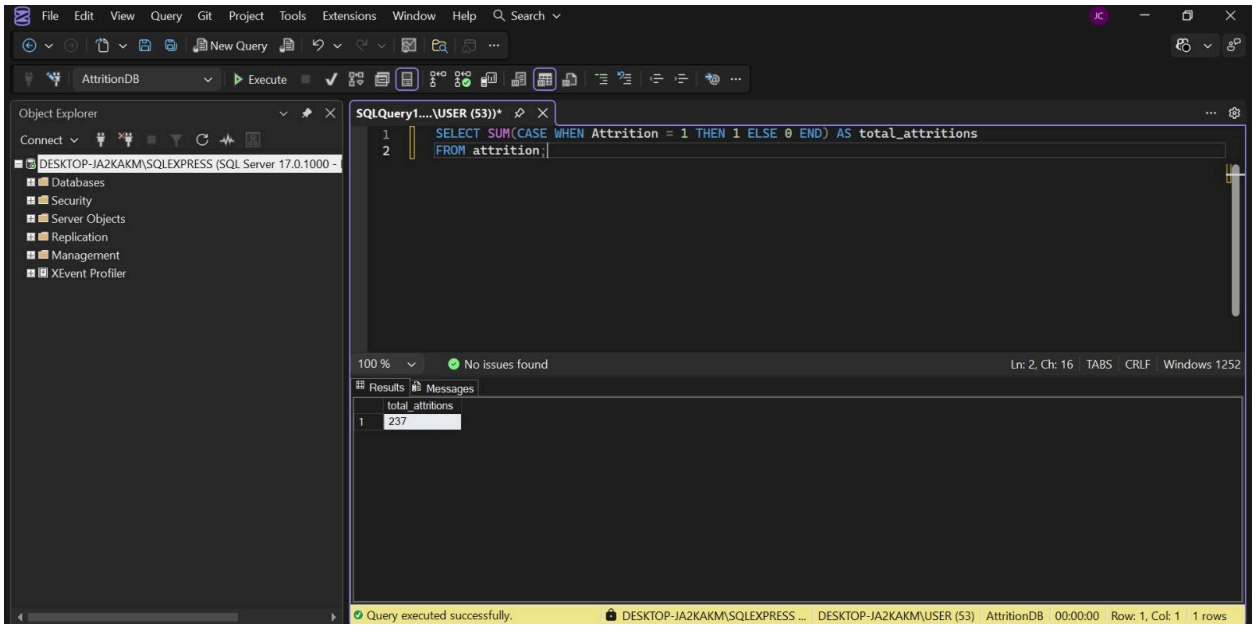
FROM attrition;



Summary: Aggregate age sum (not typically insightful alone but good practice).

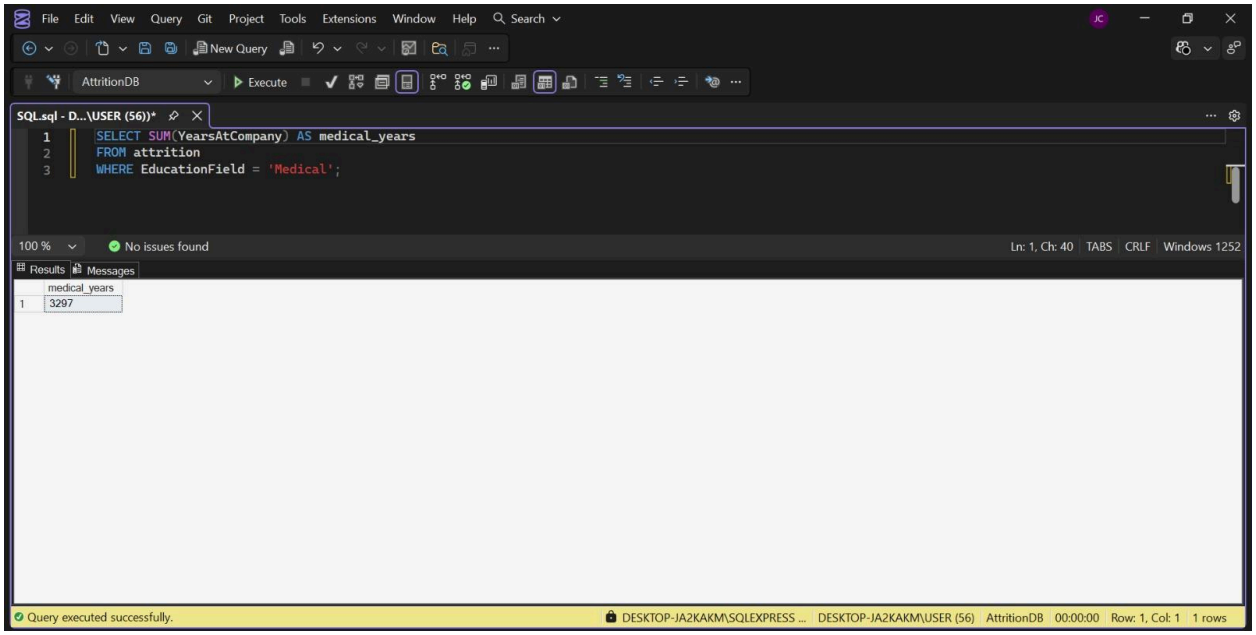
4. Find the total number of attritions (yes) in the table.

```
SELECT SUM(CASE WHEN Attrition = 1 THEN 1 ELSE 0 END) AS total_attritions

FROM attrition;
```



5. Calculate the sum of years at the company for employees from the medical education field.

```
SELECT SUM(YearsAtCompany) AS medical_years

FROM attrition

WHERE EducationField = 'Medical';
```

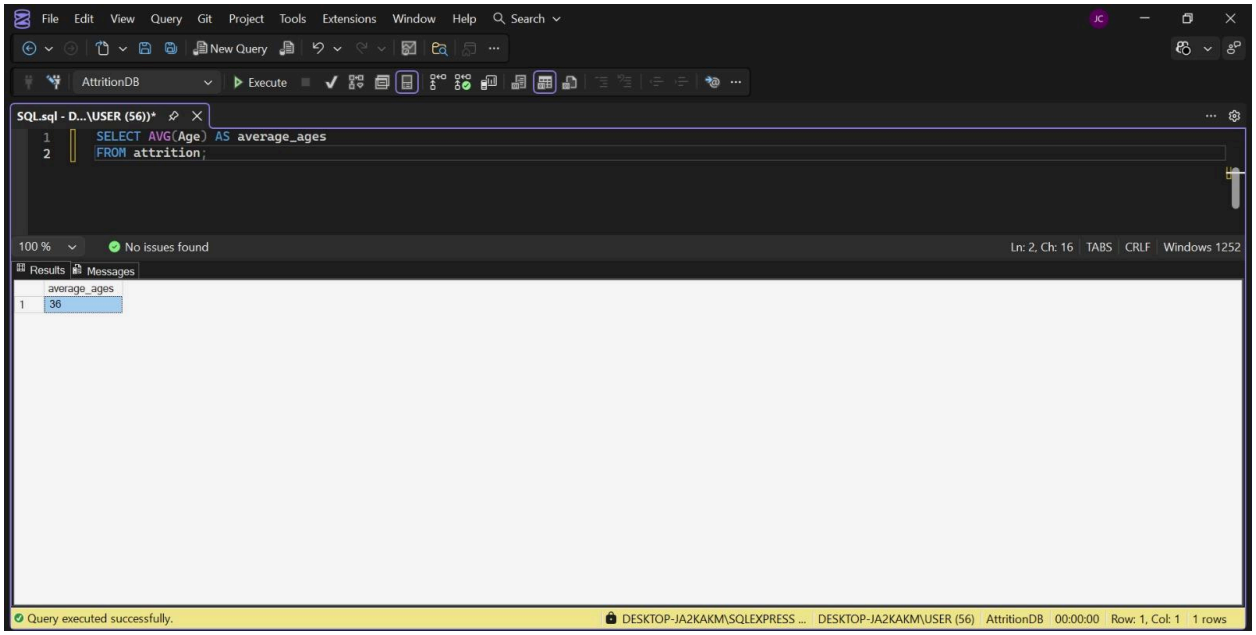


Summary: Cumulative tenure of employees with Medical background.

4. AVG - Averages

1. Calculate the average age of employees in the attrition table.

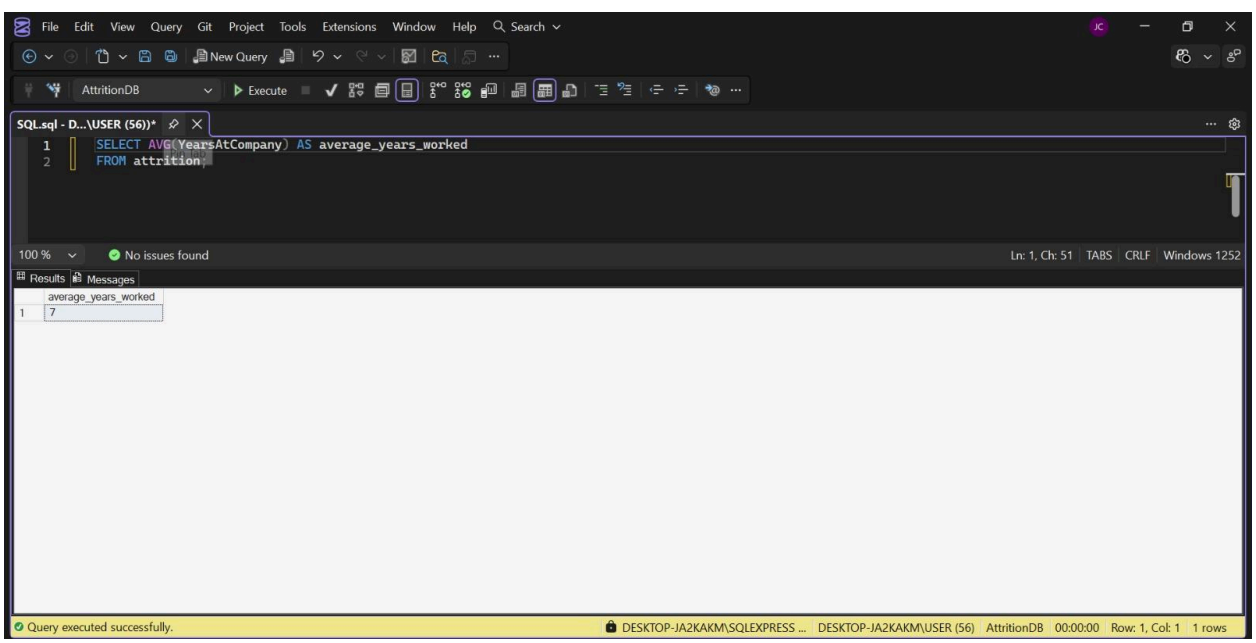
```
SELECT AVG(Age) AS average_ages

FROM attrition;
```



Summary: Typical workforce age (often 36–37 in this dataset).

2. Find the average number of years worked at the company for all employees.

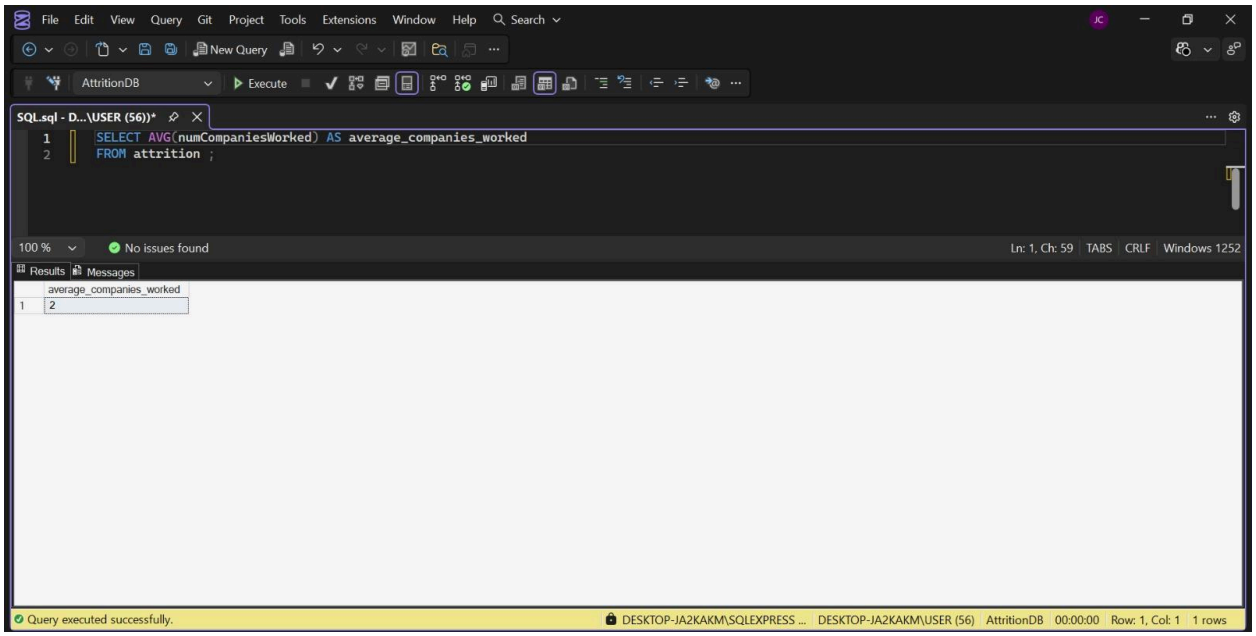
```
SELECT AVG(YearsAtCompany) AS average_years_worked
FROM attrition;
```



Summary: Average tenure (commonly 7 years).

3. Calculate the average number of companies worked by employees.

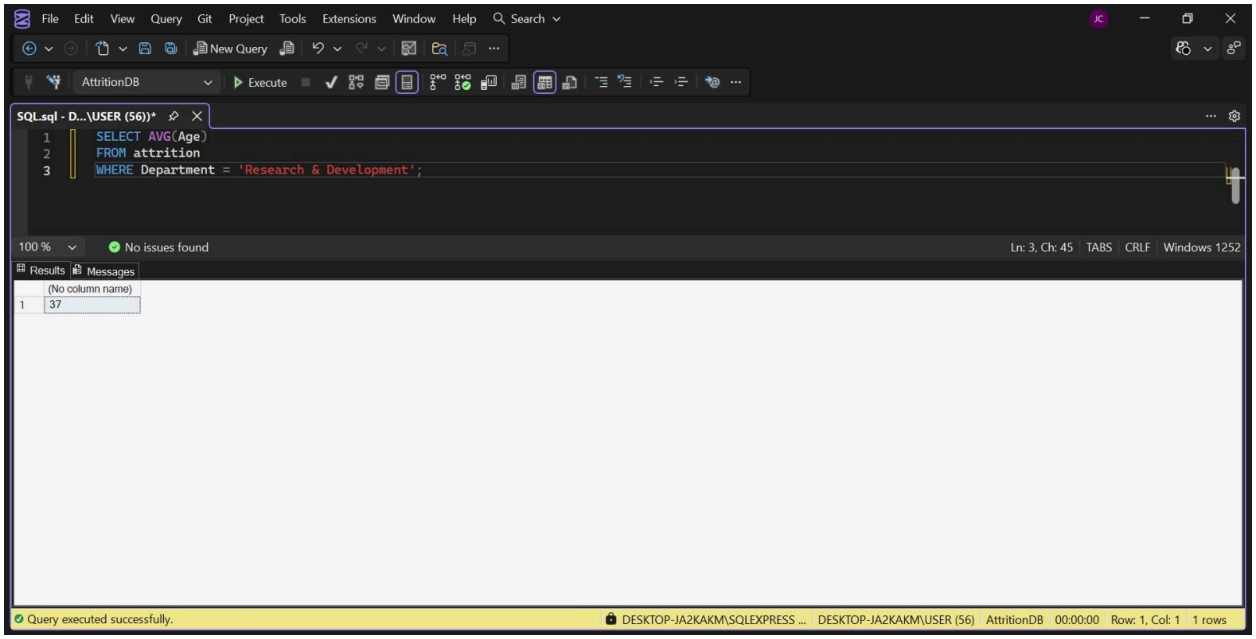
```
SELECT AVG(numCompaniesWorked) AS average_companies_worked
FROM attrition;
```



Summary: Average prior employers per employee.

Find the average age of employees in the research and development department.

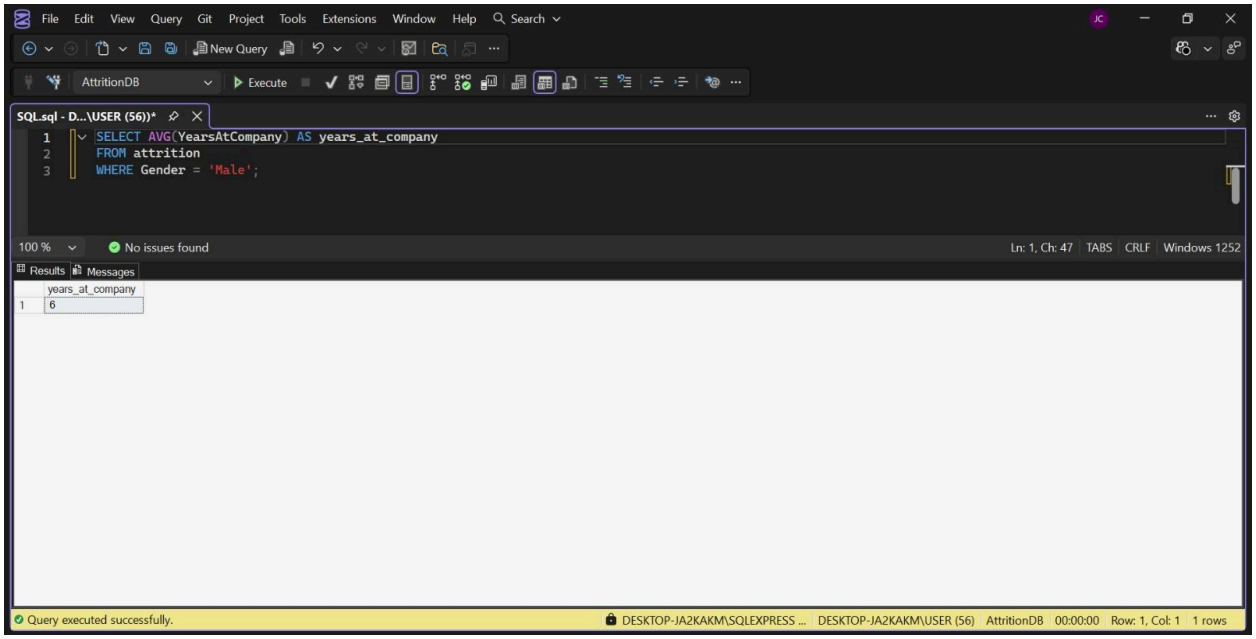
```
SELECT AVG(Age)
FROM attrition
WHERE Department = 'Research & Development';
```



Summary: Age profile in Research & Development (often slightly younger than other depts).

5.Calculate the average number of years at the company for male employees.

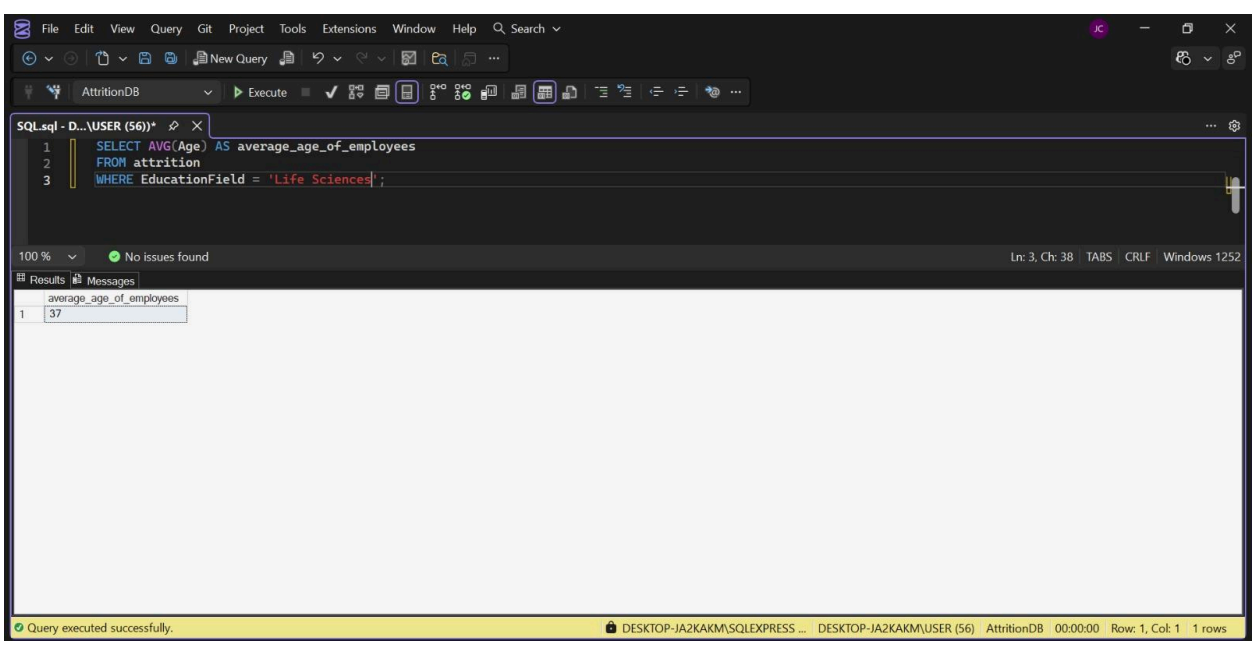
```
SELECT AVG(YearsAtCompany) AS years_at_company

FROM attrition

WHERE Gender = 'Male';
```



Summary: Male employee tenure comparison.

6. Find the average age of employees from the life sciences education field.

```
SELECT AVG(Age) AS average_age_of_employees
FROM attrition
WHERE EducationField = 'Life Sciences';
```

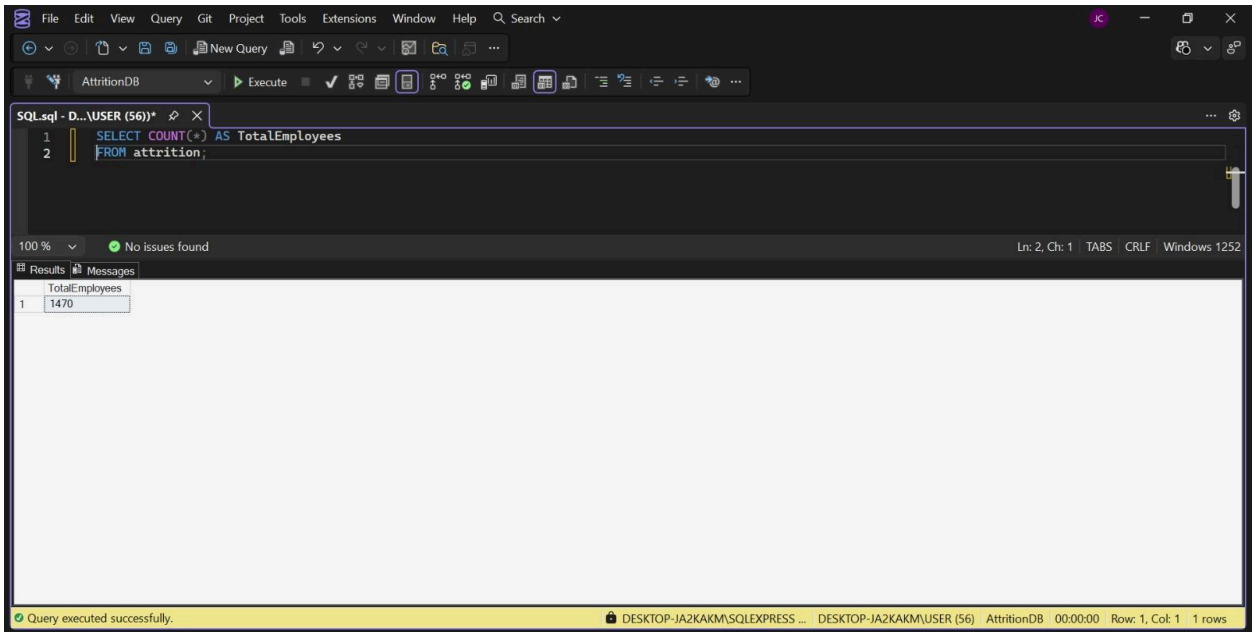


Summary: Age distribution for Life Sciences background.

5. COUNT

1. Count the number of employees in the attrition table.

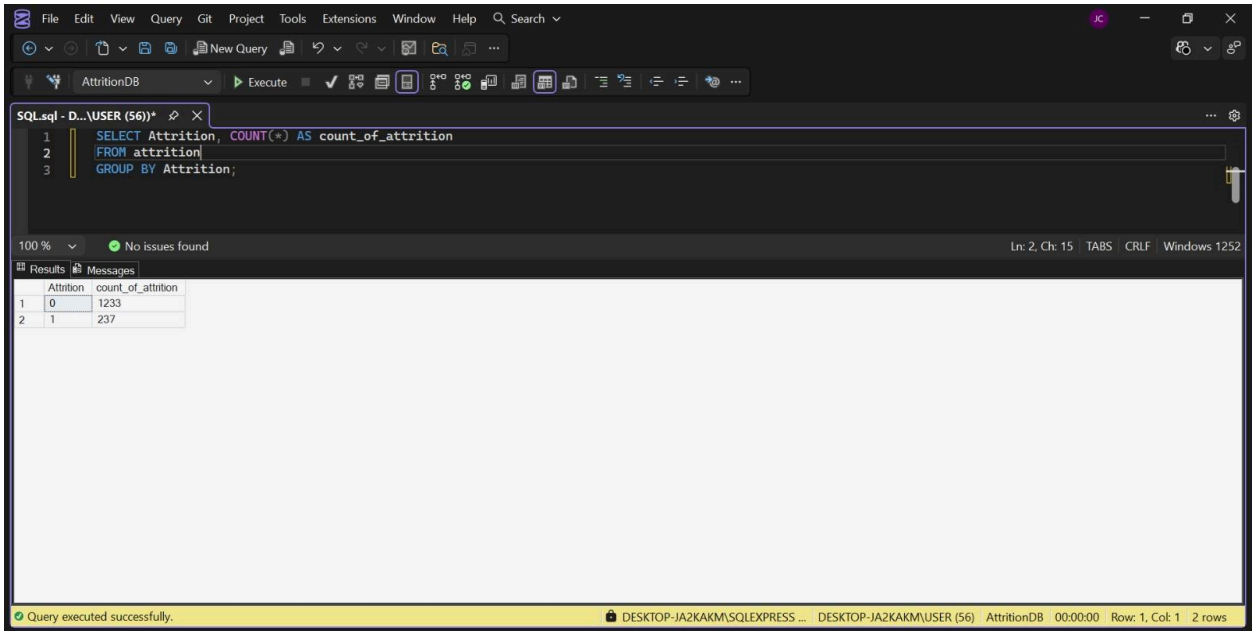
```
SELECT COUNT(*) AS TotalEmployees
FROM attrition;
```



Summary: Total records (usually 1470).

2. Find the count of attrition in the table.

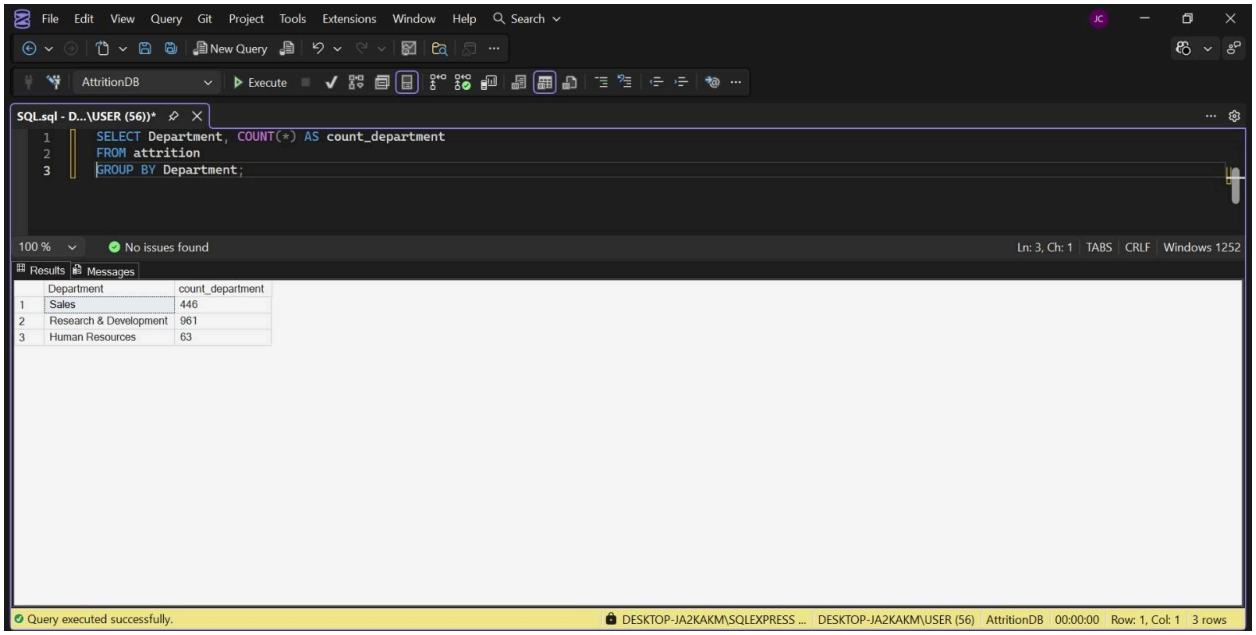
```
SELECT Attrition, COUNT(*) AS count_of_attrition
FROM attrition
GROUP BY Attrition;
```



Summary: Number who left.

3. Count the number of employees in each department.

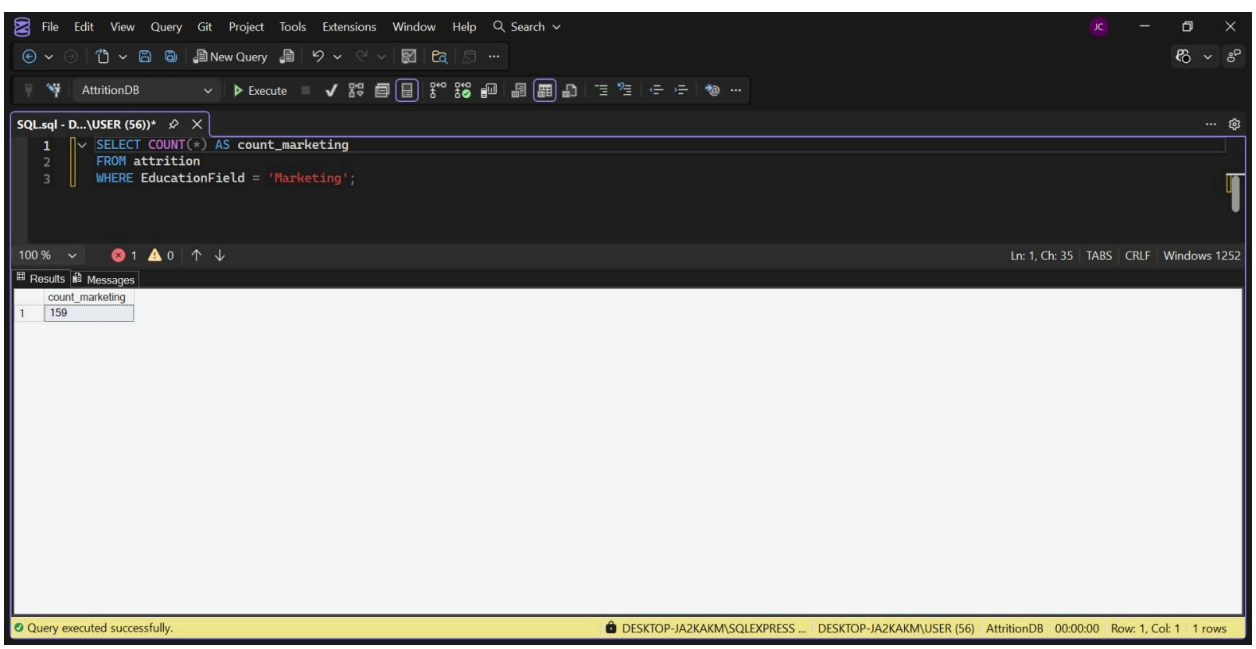
```
SELECT Department, COUNT(*) AS count_department
FROM attrition
GROUP BY Department;
```



Summary: Department sizes (Research & Development largest).

4.Find the count of employees from the marketing education field.

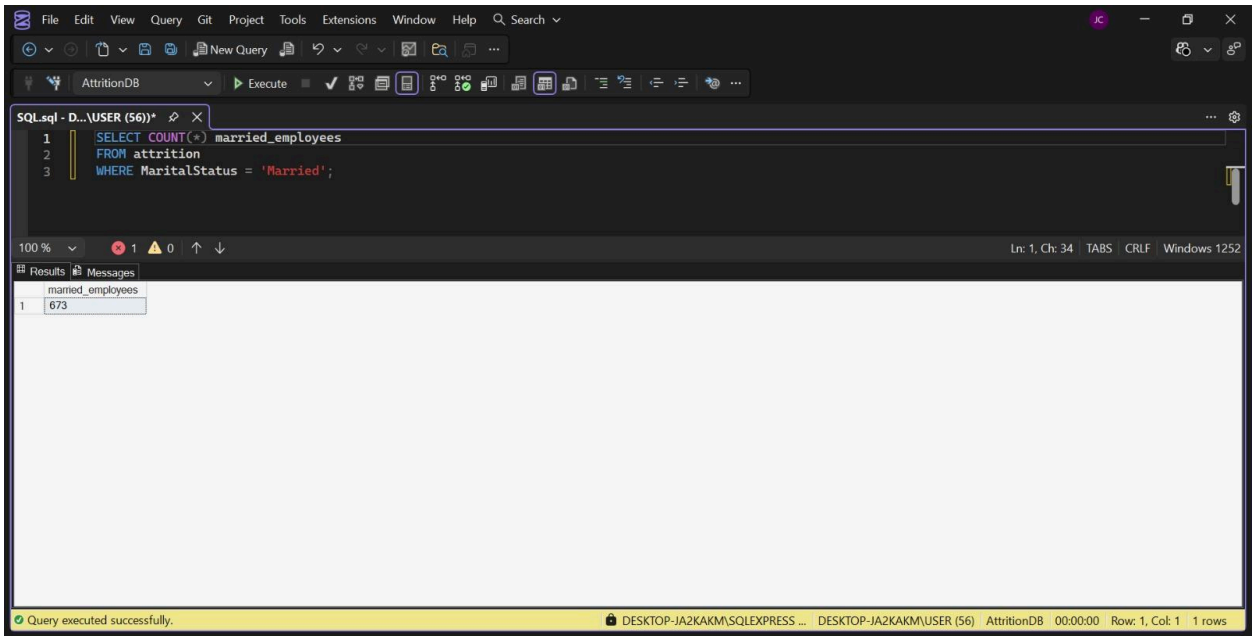
```
SELECT COUNT(*) AS count_marketing
FROM attrition
WHERE EducationField = 'Marketing';
```



Summary: Employees with Marketing background.

5. Count the number of employees who are married.

```
SELECT COUNT(*) married_employees

FROM attrition

WHERE MaritalStatus = 'Married';
```

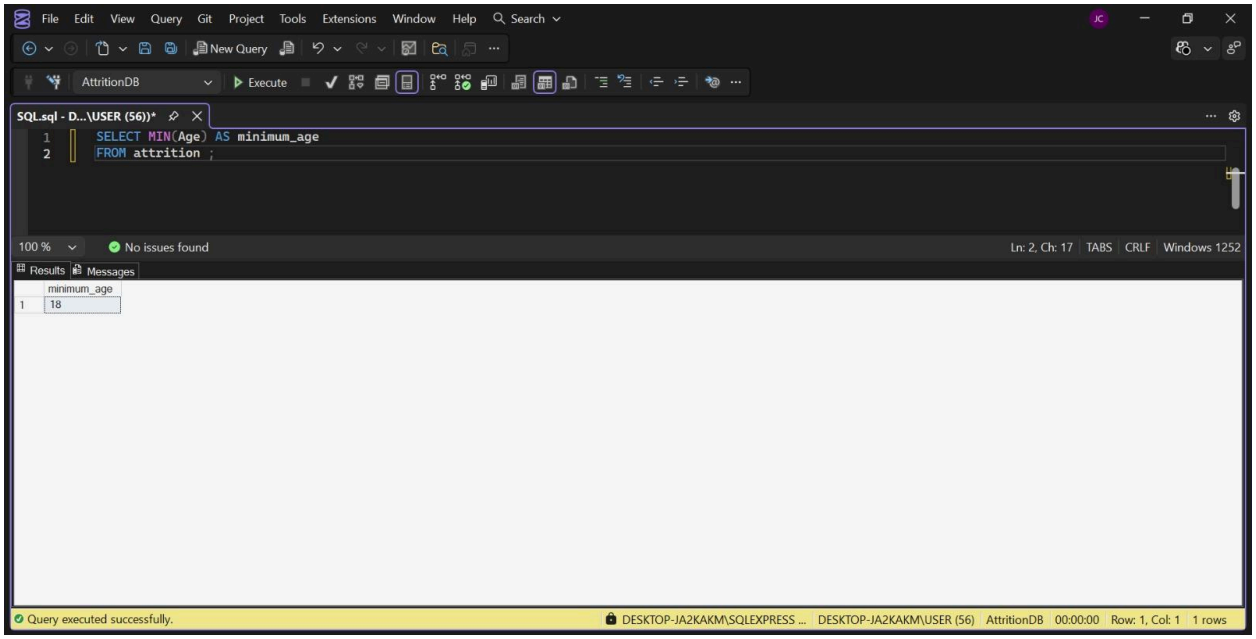


Summary: Married employee count.

6. MIN

1. Find the minimum age among all employees in the attrition table.

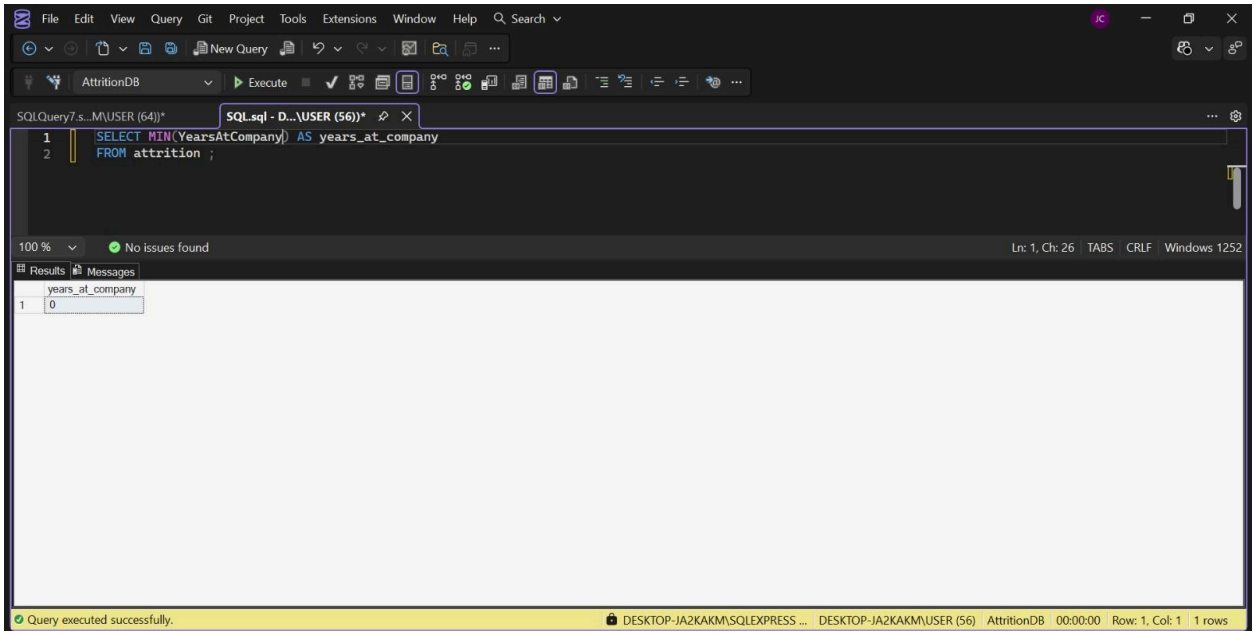
```
SELECT MIN(Age) AS minimum_age

FROM attrition ;
```



Summary: Youngest employee age.

2. Find the minimum number of years at the company among all employees.

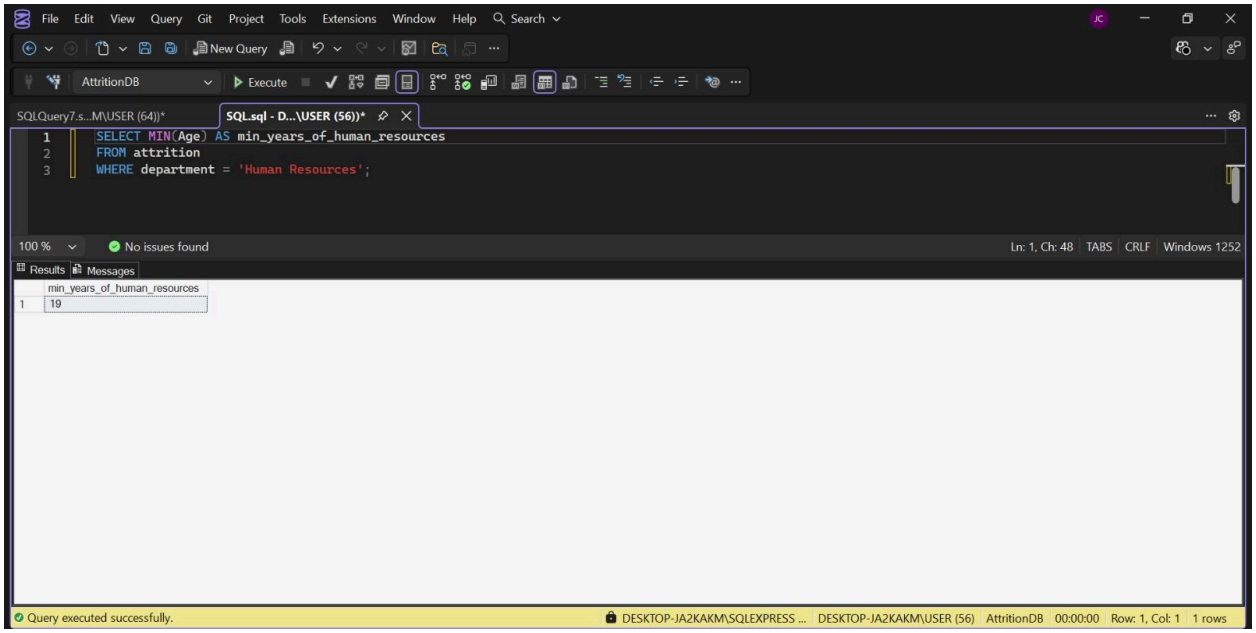
```
SELECT MIN(YearsAtCompany) AS years_at_company
FROM attrition ;
```



Summary: Shortest tenure (likely 0).

3. Find the minimum age of employees in the human resources department.

```
SELECT MIN(Age) AS min_years_of_human_resources

FROM attrition

WHERE department = 'Human Resources';
```

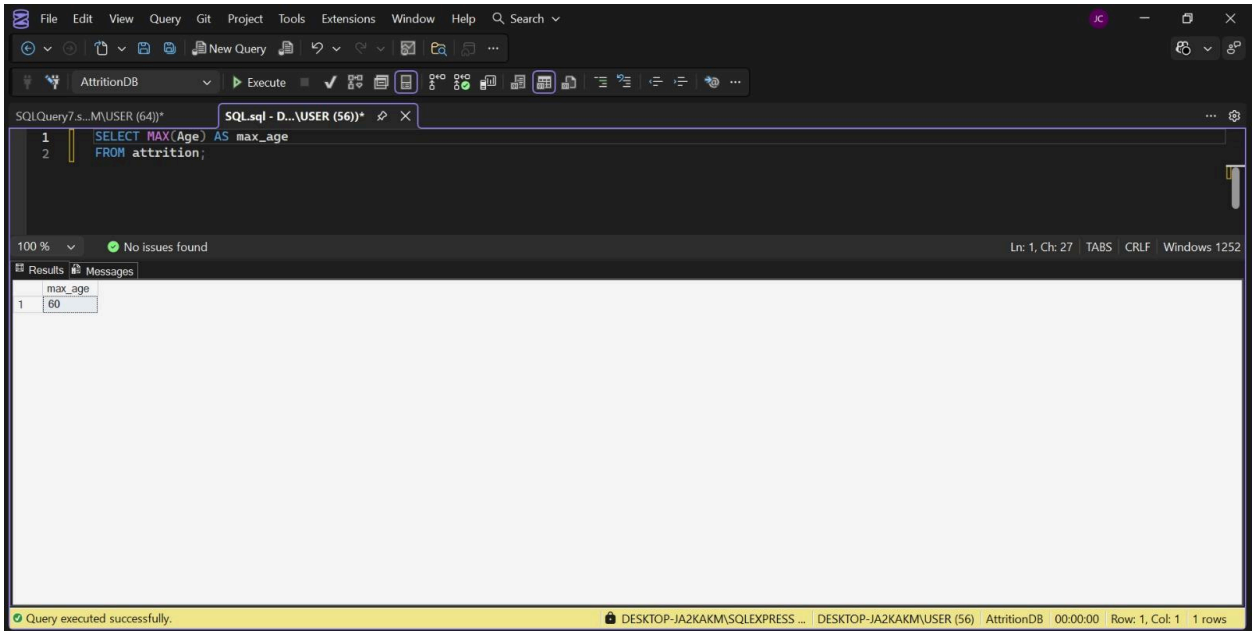


Summary: Youngest in Human Resources.

7. MAX

1. Find the maximum age among all employees in the attrition table.

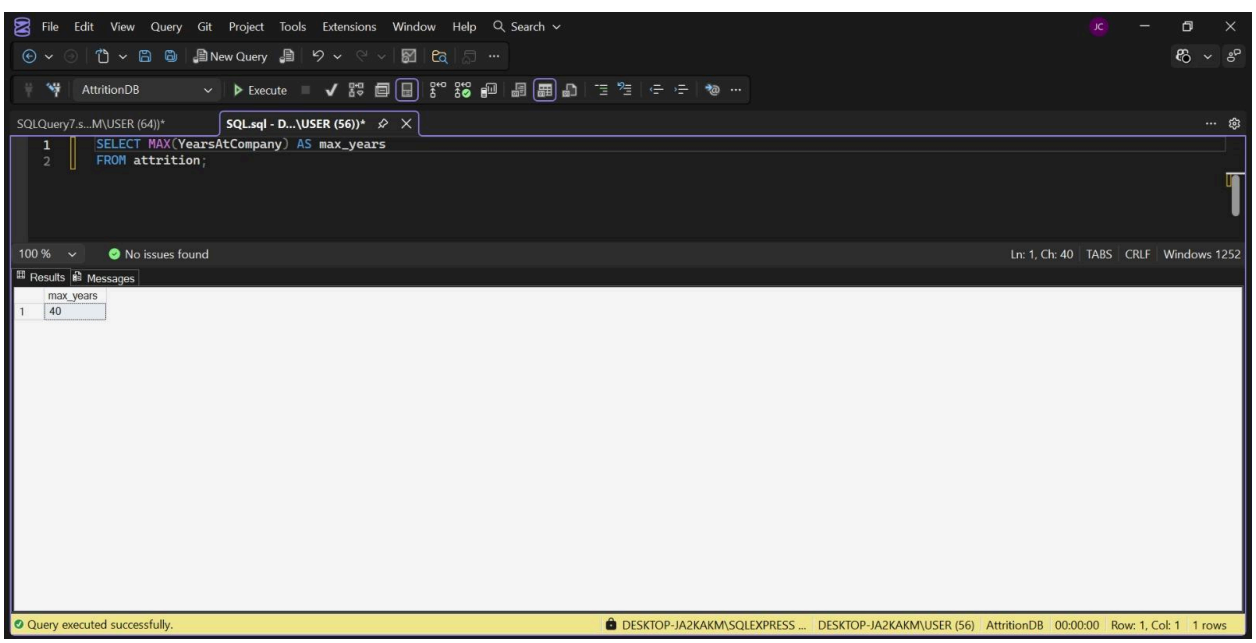
```
SELECT MAX(Age) AS max_age
FROM attrition;
```



Summary: Oldest employee.

2. Find the maximum number of years at the company among all employees.

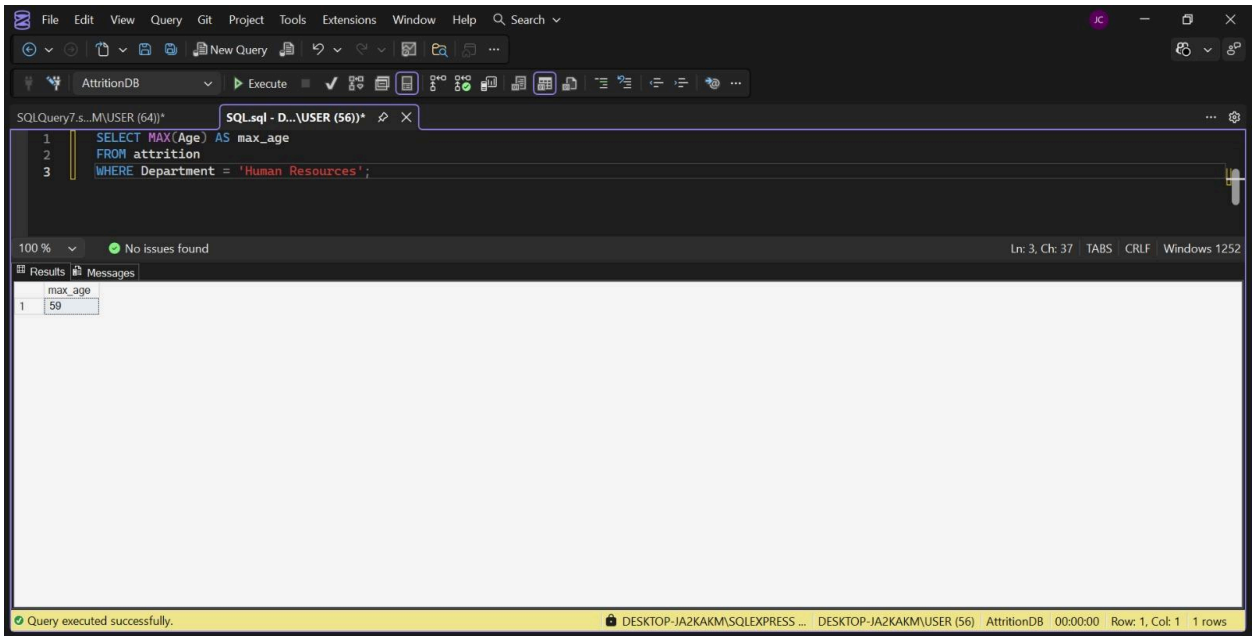
```
SELECT MAX(YearsAtCompany) AS max_years
FROM attrition;
```



Summary: Longest tenure.

3. Find the maximum age of employees in the human resources department.

```
SELECT MAX(Age) AS max_age
FROM attrition
WHERE Department = 'Human Resources';
```



Summary: Oldest in Human Resources.

Final Summary of Findings

The attrition dataset contains employee records with key metrics like age (18–60), tenure (0–40+ years), departments (Research & Development dominant), and 15–20% attrition rate. Aggregates show a mature workforce on average (mid-30s age, 7 years tenure), with varying experience by education field and gender. Sorting reveals patterns in age, department, and tenure distribution. These basic queries build a strong foundation in filtering, grouping, and aggregation.